

National Institute of Technology Kurukshetra
[B.Tech. (EE-Branch) II- Semester
Mid-Semester-Exam-II

[Advanced Engineering Physics (AEP); Course No. PHIC104]

Note: All questions are compulsory

Time allotted: 50 min.

Max. Marks: 15

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| 1. | Starting from Maxwell's equations, derive the expression for the velocity of electromagnetic waves in free space and show that: $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$ | [5] |
| 2. | Derive the following conditions:
(a) Tangential component of electric field is continuous
(b) Normal component of magnetic flux density | [3] |
| 3. | A toroid has 800 turns and carries a current of 3 A. The mean radius is 0.25 m. Using Ampere's Circuital Law, find the magnetic field inside the toroid. | [2] |
| 4. | Why are ground waves more effective at low frequencies? Explain how the conductivity of the Earth affects their propagation. | [2] |
| 5. | Prove that susceptibility of superconductor is -1 and relative permeability is zero. List any five applications of superconductivity in practical fields. | [3] |